http://www.mjpruonline.com

Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21311

F–411

B.C.A. (Third Semester) Examination, 2018

(New Course)

Paper-BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) What is Truncation Error ? Explain.

   (b) Find the absolute error in $w = xy + 2$ if $x = 2.35$, $y = 6.74$ and $z = 3.45$.

2. (a) Find a positive real root of $x \log_{10} x = 1.2$ using the bisection method.

   (b) Find a real root of the equation $x^3 - 2x - 5 = 0$ by the method of false position correct to three decimal places.

3. (a) Find a real root of $2x - \log_{10} x = 7$ correct to four decimal places using iteration method.

http://www.mjpruonline.com

http://www.mjpruonline.com

http://www.mjpruonline.com

http://www.mjpruonline.com

http://www.mjpruonline.com

(b) Find out the positive root of $x^4 - x = 10$ correct to three decimal places using Newton-Raphson method.

4. (a) From the following table, estimate the number of students who obtained marks between 40 and 45 :

Marks	No. of Students
30—40	31
40—50	42
50—60	51
60—70	35
70—80	31

(b) Find $f(22)$ from the Gauss forward formula :

x	$f(x)$
20	354
25	332
30	291
35	260
40	231
45	204

http://www.mjpruonline.com

http://www.mjpruonline.com

5. (a) Given the values :

x	$f(x)$
5	150
7	392
11	1452
13	2366
17	5202

evaluate $f(9)$, using Lagrange's formula.

(b) Find the missing term in the following table using interpolation :

x	y
0	1
1	3
2	9
3	—
4	81

6. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using :

(i) Trapezoidal rule

(ii) Simpson's 1/3 rule

(iii) Simpson's 3/8 rule

(iv) Weedle's rule and compare the results with its actual value.

http://www.mjpruonline.com

http://www.mjpruonline.com

7. (a) Find the value of y for $x = 0.1$ by Picard's method, given that :

$$\frac{dy}{dx} = \frac{y-x}{y+x}, \quad y(0) = 1$$

(b) Solve $y' = x + y$, $y(0) = 1$ by Taylor's series method; Hence find the values of y at $x = 0.1$ and $x = 0.2$.

8. (a) Apply Runge-Kutta fourth order method to find an approximate value of y when $x = 0.2$ given that $dy/dx = x + y$ and $y = 1$ when $x = 0$.

(b) Explain the difference between linear and non-linear regression with proper example.

http://www.mjpruonline.com
Whatsapp @ 9300930012
Your old paper & get 10/-
पुराने पेपर्स भेजे और 10 रुपये पायें,
Paytm or Google Pay से

http://www.mjpruonline.com